

Sales Capstone Project

Short Summary – Findings & Recommendations

Total Sales	Total Orders	Units Sold	Avg. Order Sales
971,956.32	6,645	20,038	146.27

Key Findings

- **Regional performance:** North generated the highest total sales (309,130.94).
- **Category performance:** Home & Kitchen was the leading category (385,814.40).
- **Product performance:** Vacuum Cleaner was the top product by total sales (166,881.86).
- **Channel performance:** Online generated the highest sales (520,477.70).
- **Sales drivers:** Units sold and unit price show the strongest positive relationship with sales among the main numeric variables.
- **Data quality:** 132 MarketingSpend values were missing and should be handled before analysis/model deployment.

Regression Model

A Gradient Boosting Regression model was developed using product, region, category, customer segment, channel, units sold, price, discount, marketing spend and time-based features. The model achieved an R^2 of 0.442, with MAE of 12.37 and RMSE of 161.71 on the test set. The relatively high RMSE compared with MAE suggests that a small number of large-error/outlier transactions may be affecting model performance.

Recommendations

- Prioritize high-performing categories and products in inventory, promotions and sales planning.
- Use the strongest region and channel as benchmarks while investigating underperforming areas.
- Analyze discounts by product and channel rather than increasing discounts broadly.
- Measure marketing ROI at campaign/channel level instead of relying only on total marketing spend.
- Review unusually large transactions and missing values before using the model for business decisions.
- Use the Power BI dashboard to monitor sales trends, regional performance, category mix, products and channels regularly.

Conclusion: The analysis provides a clear view of sales performance and identifies the main areas for business focus. Combining the regression model with the Power BI dashboard can support more data-driven sales planning and performance monitoring.